

State-Conditioned Marginal Utility for Multi-Context Prediction

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Abstract—Auxiliary contexts do not have fixed predictive value: their contribution changes as other contexts are selected. We formulate multi-context selection through set-conditioned marginal utility, the reduction in end-task loss produced by adding a candidate to the current context set. The formulation separates structural opportunity from estimation error and applies beyond a single predictor or task family. We introduce Response-Aware Marginal-Utility Routing, which narrows the candidate pool with cached representations and reranks retained candidates using compact predictor-response summaries and a utility-aware ranking objective. A decision analysis links marginal-utility estimation error to routing regret, and a general smooth-loss view relates predictor response to marginal value, with exact and two-sided statements for Bregman and softmax-cross-entropy losses. The analysis also predicts when the approach pays off: because the identifiable term is the residual of the fixed predictor itself, stronger predictors leave less realizable headroom even when the oracle opportunity persists. Experiments confirm both the gains and the boundary across standard forecasting benchmarks, a subset-capable nonlinear backbone and a second multi-context task family, and they separate the ranking, which transfers, from the stopping rule, which does not. The comparison is not flattering to the standard toolkit: in the regime where utility is estimable, learned marginal-utility routing is significantly ahead of the best of fourteen competitors on three benchmarks, unresolved on two and below simple pooling on the sixth, while relevance ranking and maximum marginal relevance are significantly worse than using no auxiliary context at all on four of the six and the strongest competitor on every benchmark is the trivial policy of including every candidate; in the regime where it is not, no method separates from simply using every context, and the sign of the retrieval heuristic is predicted by a single label-free statistic that also yields a label-free adaptive rule. Two controlled sweeps locate the mechanism rather than merely reporting the boundary: adapting the fixed predictor for two epochs removes about five sixths of the value routing can realize while leaving the oracle opportunity untouched, and enlarging the candidate library from sixteen to one hundred and twenty-eight raises that opportunity by forty percent while driving the router to the level of random selection. Removing the shortlist entirely changes the result by an amount indistinguishable from zero, which locates the limit in what the predictor’s response can reveal rather than in the efficiency compromise that observes it. The measurement protocol is then validated as a deployment decision: on a labelled pilot of twenty-four targets it reproduces the full-deployment verdict ninety-six percent of the time, with ninety-eight percent sensitivity for the decision to route at all, and splitting the pilot in half yields a self-audit that certifies a verdict with ninety-nine percent agreement.

Index Terms—multi-context prediction, marginal utility, context routing, decision-focused learning, negative transfer.

I. INTRODUCTION

Predictive systems increasingly combine multiple auxiliary contexts, including retrieved examples, historical observations,

and neighboring records. The usefulness of each context depends on what the predictor has already received. A context that improves a prediction in isolation can become redundant after similar information is selected, while another candidate with lower standalone value can correct the remaining error. A selection rule that assigns each candidate a fixed score cannot represent this change.

We model the value of a candidate directly as its marginal reduction in end-task loss,

$$m(j \mid A) = F(A \cup \{j\}) - F(A), \quad (1)$$

where A is the current context set and F measures the expected loss reduction relative to an anchor-only predictor. The marginal depends on the predictor’s current residual and on the direction and magnitude of the prediction change induced by the candidate. It therefore differs from a standalone score $u(j \mid \emptyset)$: the same candidate can be useful at one selection state and redundant or harmful at another.

This distinction is structural rather than terminological. A perfect standalone score fixes candidate values before selection. A sequential oracle recomputes the true marginal after each addition. Their gap,

$$H_{\text{state}} = \frac{G_{\text{seq}} - G_{\text{stand}}}{G_{\text{seq}}}, \quad (2)$$

measures the value created solely by conditioning on the selected set. We call this quantity structural headroom. Controlled experiments show that redundancy enlarges it, and that a learned marginal-utility router can surpass an exact standalone ranking when redundancy is high.

Learning to estimate marginal utility is harder than predicting a fixed candidate score. The router must distinguish candidates whose utilities are close, and it must know how each candidate changes the fixed predictor. As the candidate pool grows, the utility gap between the best alternatives shrinks. Cached context representations, in turn, omit information carried by the predictor’s response to a candidate. These two properties set the design requirements for a practical router.

We introduce Response-Aware Marginal-Utility Routing (R-MUR). A low-cost cached score first narrows the candidate pool. The fixed predictor is evaluated only on this shortlist, and compact summaries of its response are used to rerank the retained candidates. Training combines marginal-utility regression with a pairwise objective weighted by the utility gap, which aligns estimation with the final selection decision. The design follows directly from the two properties above: predictor-response features expose the change that enters the

marginal, and ranking calibration targets small decision margins.

The resulting evidence chain is deliberately narrow, and it has a boundary that we characterize rather than hide. Controlled studies isolate how redundancy and candidate-set growth change the selection problem. Across standard traffic benchmarks, R-MUR improves learned context selection over cached and standalone baselines, while a multi-target study shows that state-conditioning headroom persists across sensors. These results establish state-conditioned marginal utility as a practical decision object for budgeted multi-context prediction and identify predictor response as a key signal for learning it. We then test the same framework against a subset-capable nonlinear backbone and against a second task family. The nonlinear backbone preserves structural headroom but shrinks the share of it that a router can realize: response-free learned routers become indistinguishable from random selection and R-MUR’s advantage over them is no longer statistically resolved. The second family, whose candidate demonstrations are not redundant, is already solved by nearest-neighbour relevance. Together with the smooth-loss analysis, these results locate the regime in which sequential marginal routing pays off: candidates must overlap in what they contribute, and the fixed predictor must retain a residual that the router can observe.

a) Scope and companion manuscripts.: This paper reports the method and the generalisation evidence: the pipeline, the ten-method baseline matrix, the strong-backbone boundary, the second task family, the compute profile and the adaptive rule. The identifiability theory — why the predictor’s response decides whether utility can be estimated, and why stronger predictors make routing harder — is developed and proved in a companion manuscript. The summary propositions reproduced here (Section V) are stated so that this paper’s predictions can be read without it.

II. RELATED WORK

Auxiliary contexts and demonstrations adapt a frozen predictor to a query [1, 2]. Their quality and ordering matter, and many methods retrieve or select examples before prediction [3–6]. Other work studies prompt-order sensitivity and calibration [7, 8]. Retrieval-augmented systems condition predictions on external passages and records [9–11], and retrieval scale and adaptation are studied in [12–15]. Recent work combines retrieval with adaptive chains and causal reasoning [16, 17]. Diversity-aware selection penalizes repeated content or maximizes coverage [18–20], and listwise rerankers compare candidate sets directly [21–25]. Long-context studies show that position and context composition change how a model uses retrieved information [26]. These approaches determine which information is relevant or diverse; they do not directly estimate the change in end-task loss produced by adding a candidate to the context already selected.

Outcome-aware selection moves closer to the final prediction objective by valuing examples through their effect on model output. Hashimoto et al. estimate the incremental utility of a demonstration relative to an empty prompt [27]. Data

valuation and influence estimation assign credit to training examples through their effect on predictions [28–31]. These methods characterize a candidate’s standalone contribution. Our object conditions on a nonempty, evolving selected set, so the value of the same candidate can change after every routing decision.

Set-selection methods provide the structural counterpart to this decision problem. Submodular objectives formalize coverage, diversity, and diminishing returns [32–34], while determinantal point processes model diverse subsets probabilistically [35]. Adaptive submodularity and influence maximization extend these tools to sequential decisions [36, 37], and streaming or contextual variants address budgeted settings [38, 39]. Classical monotone formulations reward additional elements; context utility in our setting can also be negative because a candidate can increase prediction loss. Negative transfer has been characterized in general transfer learning [40–42] and mitigated in sequential recommendation through multi-modal re-learning [43].

Routing and decision-focused learning treat selection as a learned decision problem. Mixture-of-experts systems route inputs to expert subsets [44–46], and sparse or soft routing architectures reduce computation while increasing capacity [47–50]. Routing networks select computation paths for multi-task learning [51, 52], and recent systems apply dynamic routing across language, vision-language-action, and collaborative-perception tasks [53–55]. Decision-focused learning connects prediction quality to downstream decision quality [56–59], and task-based end-to-end learning follows the same principle [60]. Sequential selection is related to bandit exploration and off-policy value estimation [61, 62]. R-MUR applies this perspective to auxiliary contexts for a fixed predictor, where the routing target is the marginal change in end-task loss and the predictor’s response becomes part of the available decision information.

III. STATE-CONDITIONED CONTEXT SELECTION

Let S_0 be an anchor context, x a query, y its target, and $\mathcal{C} = \{C_1, \dots, C_K\}$ a candidate pool. A selected set is $A \subseteq \{1, \dots, K\}$. A predictor produces $f_A(x)$ from S_0 and the contexts indexed by A ; the predictor is trained before routing and kept fixed during context selection and router training.

We define the net set utility as

$$F(A) = \mathbb{E} [\ell(f_{\emptyset}(x), y) - \ell(f_A(x), y)] - \lambda c(A), \quad (3)$$

where $c(A)$ is a context cost. The marginal utility of a candidate is

$$m(j \mid A) = F(A \cup \{j\}) - F(A). \quad (4)$$

For one observed episode, the training target is the counterfactual loss difference

$$\tilde{m}(j \mid A) = \ell(f_A(x), y) - \ell(f_{A \cup \{j\}}(x), y) - \lambda c_j. \quad (5)$$

This is a realized observation of the marginal. Its interpretation as an expectation requires a fixed predictor and a specified state-sampling rule.

For squared loss, let $\mu = \mathbb{E}[y \mid S_0, x, \mathcal{C}]$ and let $d_{A,j} = f_{A \cup \{j\}}(x) - f_A(x)$. The marginal separates into

$$m(j \mid A) = 2d_{A,j}(\mu - f_A(x)) - d_{A,j}^2 - \lambda c_j. \quad (6)$$

The first term is the interaction between the current residual and the direction of the candidate-induced correction, the second term penalizes overcorrection, and the third accounts for cost. Relevance and standalone candidate scores do not determine these three terms, and the same candidate can be useful at one state and harmful at another.

For a smooth loss $\ell(f, y)$ and prediction change $d_{A,j} = f_{A \cup \{j\}}(x) - f_A(x)$, a second-order expansion gives

$$\ell(f_A, y) - \ell(f_{A \cup \{j\}}, y) = -\nabla_f \ell(f_A, y)^\top d_{A,j} - \frac{1}{2} d_{A,j}^\top \nabla_f^2 \ell(\xi, y) d_{A,j}, \quad (7)$$

for some ξ on the segment between f_A and $f_{A \cup \{j\}}$. Taking expectations yields

$$m(j \mid A) = -\mathbb{E}[\nabla_f \ell(f_A, y)^\top d_{A,j}] + O(\mathbb{E}\|d_{A,j}\|^2) - \lambda c_j. \quad (8)$$

When the Hessian is bounded, the second-order remainder is controlled by the response magnitude. Section V turns Equation (7) into two-sided bounds with explicit constants, gives an exact identity for Bregman losses that contains Equation (6) as a special case, and shows that response features are the correct sufficient statistic for the squared-loss case. Predictor response therefore carries information about marginal utility beyond quadratic losses.

Two oracle policies expose the structural value of this state dependence. The standalone oracle ranks candidates by the true standalone marginal $m(j \mid \emptyset)$ and keeps that order fixed. The sequential oracle recomputes the true marginal after every addition. Let G_{stand} and G_{seq} denote their prediction gains over the anchor-only predictor. We measure structural headroom as

$$H_{\text{state}} = \frac{G_{\text{seq}} - G_{\text{stand}}}{G_{\text{seq}}}. \quad (9)$$

This quantity is an evaluation diagnostic. It isolates the value created by conditioning on the selected set, independently of router estimation error.

IV. RESPONSE-AWARE MARGINAL-UTILITY ROUTING

R-MUR estimates marginal utility with the fixed predictor. Every candidate is first represented by cached context features. At state A , let $\hat{m}_C(j \mid A)$ denote a cached-only utility score. A shortlist is

$$\mathcal{Q}_A = \text{TopQ}_{j \notin A} \hat{m}_C(j \mid A), \quad |\mathcal{Q}_A| = q \ll K. \quad (10)$$

The shortlist reduces the number of predictor evaluations while retaining the candidates that are most promising under the cached representation.

For each retained candidate, R-MUR computes a compact response summary from the fixed predictor,

$$r_{A,j} = \psi(f_A(x), f_{A \cup \{j\}}(x)). \quad (11)$$

The summary contains the level of the current prediction, the level after adding the candidate, the mean and standard

deviation of the response change, its mean absolute value, its normalized Euclidean norm, and its maximum absolute value. These statistics describe how the candidate changes the predictor, not merely how similar the candidate is to the query.

The response-aware utility model scores retained candidates with

$$\hat{m}_R(j \mid A) = g(h_Q, h_A, h_j, r_{A,j}), \quad j \in \mathcal{Q}_A, \quad (12)$$

where h_Q , h_A , and h_j are cached representations of the query, the selected set, and the candidate. The set encoder is permutation invariant. Training minimizes

$$\mathcal{L} = \mathcal{L}_{\text{Huber}} + \beta \sum_{i,j: \hat{m}_i > \hat{m}_j} |\tilde{m}_i - \tilde{m}_j| \log(1 + \exp[-(\hat{m}_i - \hat{m}_j)]), \quad (13)$$

where the pairwise term is evaluated within the same selection state. The gap weight gives larger influence to pairs that can change the argmax decision. The regression term keeps the utility scale calibrated on states without a dominant alternative.

At inference, R-MUR starts from $A_0 = \emptyset$ and repeats the following steps until the budget is reached or a stopping condition is met. It scores all remaining candidates with the cached model, forms $\mathcal{Q}_A$, computes response summaries for the shortlist, reranks the shortlisted candidates with $\hat{m}_R$, and adds the highest-scoring positive candidate. The selected set is updated and the process repeats. The procedure is greedy; it targets one-step decision quality under a context budget.

A calibration-derived threshold controls when routing stops. On a held-out split, let r_i be the absolute residual between predicted and observed marginals. For level α , the radius $q_\alpha = \text{Quantile}_{1-\alpha}(\{r_i\})$ defines a symmetric interval

$$[L_{A,j}, U_{A,j}] = [\hat{m}_R(j \mid A) - q_\alpha, \hat{m}_R(j \mid A) + q_\alpha]. \quad (14)$$

This threshold changes only the stopping decision; the candidate ranking remains the response-aware score.

A. A head shaped by the exact identity

The experiments in Section VI-E show that a generic scalar head over concatenated state, candidate and response features does not recover marginal utility when the fixed predictor is strong. The analysis in Section V says why, and prescribes the functional form. Conditional on the state, Equation (21) gives

$$\mathbb{E}[\tilde{m}(j \mid A) \mid x, A] = \langle g_A(x), d_{A,j}(x) \rangle - \|d_{A,j}(x)\|^2 - \lambda c_j, \quad g_A(x) = 2 \quad (15)$$

Only g_A has to be learned: the response $d_{A,j}$ is observed once the predictor has been probed, and the curvature penalty is an exact function of that same response. We therefore replace the scalar head by a *bilinear response head* that emits a vector in prediction space and scores

$$\hat{m}_B(j \mid A) = \langle g_\theta(\phi_A(x), c_j), d_{A,j}(x) \rangle - \|d_{A,j}(x)\|^2 - \lambda c_j, \quad (16)$$

where $\phi_A(x)$ collects the same state and candidate features as before and g_θ is a small network. This is the minimal-capacity model consistent with the identity: it cannot represent a score

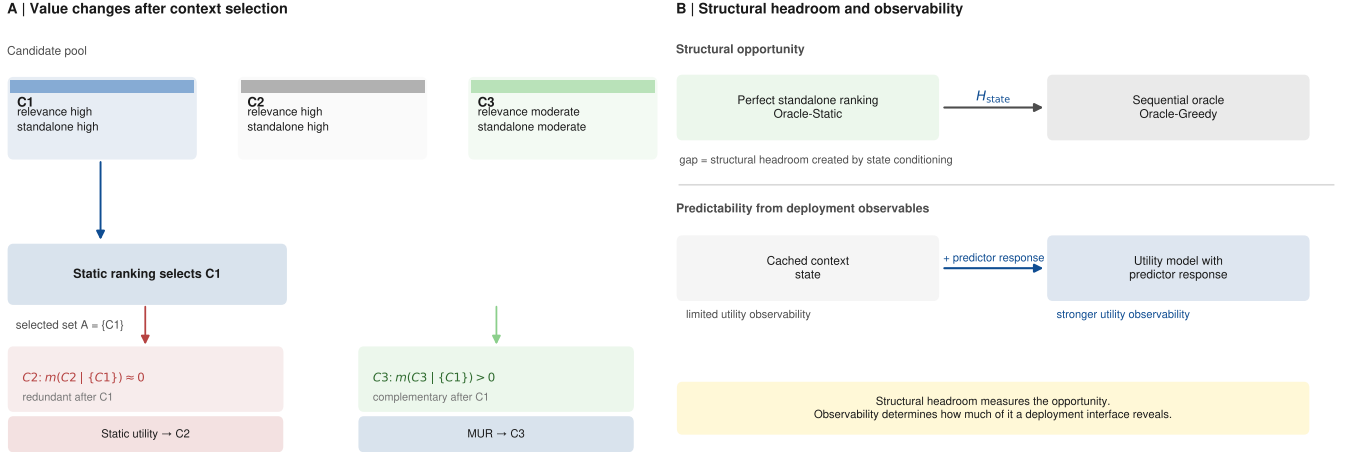


Fig. 1. State-conditioned context selection. A candidate’s standalone value can change after similar information is selected. The gap between a standalone oracle and a sequential oracle measures structural headroom, while the information available at inference time determines how much of that value a router can estimate.

that ignores the response, it never has to learn the curvature term, and the only statistical task left is estimating the state-dependent coefficient. The training target is unchanged (the realised marginal), so the head is trained with the same regression and gap-weighted ranking objective as before; the difference is purely inductive bias. For a general smooth loss the same construction uses the identity of Proposition 2 in place of the squared-loss form, and the cross-entropy bound of Proposition 3 replaces the exact penalty by its upper bound.

V. ANALYSIS

A. Predictor response and marginal utility

The router observes how a candidate changes the frozen predictor’s output. The following results quantify how much that response tells us about the candidate’s marginal utility. Proofs are in Appendix A.

Proposition 1 (Response expansion). *Assume that, for every episode, $f \mapsto \ell(f, y)$ is twice continuously differentiable on the segment between $f_A(x)$ and $f_{A \cup \{j\}}(x)$, and that $\|\nabla^2 \ell(z, y)\|_2 \leq L$ on that segment. With $d_{A,j}(x) = f_{A \cup \{j\}}(x) - f_A(x)$,*

$$m(j | A) = -\mathbb{E}[\nabla \ell(f_A, y)^\top d_{A,j}] \quad (17)$$

$$|m(j | A) + \mathbb{E}[\nabla \ell(f_A, y)^\top d_{A,j}] + \lambda c_j| \leq \frac{L}{2} \mathbb{E}\|d_{A,j}\|^2. \quad (18)$$

The leading term is the alignment between the loss gradient at the current prediction and the direction in which the candidate moves the predictor. The remainder is second order in the response magnitude, so the first-order term is accurate exactly when the candidate perturbs the predictor little, and a large response can cost at most $(L/2)\mathbb{E}\|d_{A,j}\|^2$.

Proposition 2 (Exact identity for Bregman losses). *Let $\ell(f, y) = D_\phi(y, f) = \phi(y) - \phi(f) - \nabla \phi(f)^\top (y - f)$ for convex differentiable ϕ . Then*

$$m(j | A) = \mathbb{E}[(\nabla \phi(f_{A \cup \{j\}}) - \nabla \phi(f_A))^\top (y - f_A)] - \mathbb{E}D_\phi(f_A, f_{A \cup \{j\}}). \quad (19)$$

Equation (19) is exact and requires no smoothness constant. The second term is a non-positive movement penalty in the Bregman geometry. With $\phi(z) = \|z\|^2$ it reduces to the squared-loss identity of Equation (6). If ϕ is μ -strongly convex and L -smooth, the penalty is bounded by $(\mu/2)\|d\|^2$ and $(L/2)\|d\|^2$ from below and above.

Proposition 3 (Cross-entropy, explicit constants). *For logits $f \in \mathbb{R}^C$, one-hot y , and softmax cross-entropy $\ell(f, y) = -f_y + \log \sum_k e^{f_k}$ with $p = \text{softmax}(f)$, the Hessian $\nabla^2 \ell = \text{diag}(p) - pp^\top$ is positive semidefinite with spectral norm at most $1/2$. Consequently*

$$\langle e_y - p_A, d_{A,j} \rangle - \frac{1}{4} \mathbb{E}\|d_{A,j}\|^2 - \lambda c_j \leq m(j | A) \leq \langle e_y - p_A, d_{A,j} \rangle - \lambda c_j. \quad (20)$$

The upper bound holds because the second-order term penalises any positive semidefinite curvature; the lower bound says that a large response can lose at most a quarter of its squared size in logit space.

Proposition 4 ((Response) sufficiency for squared loss). *For squared loss, conditioning on the state gives*

$$\mathbb{E}[\tilde{m}(j | A) | x, A] = \langle g_A(x), d_{A,j}(x) \rangle - \|d_{A,j}(x)\|^2 - \lambda c_j, \quad g_A(x) = 2(\mu \quad (21)$$

where $\mu(x) = \mathbb{E}[y | x, A]$.

Thus, conditional on the state, the Bayes-optimal marginal-utility predictor is affine in the response vector plus a known quadratic penalty. This is the parameterisation used by the response-aware router: state features supply the coefficient $g_A(x)$, response features supply $d_{A,j}$, and the quadratic term is a function of the response alone. Any score that is independent of the response can only recover the state average and provably

omits the $\langle g_A, d_{A,j} \rangle$ interaction that the headroom diagnostic of Equation (9) measures. For a general differentiable loss the same argument replaces $2(f_A - y)$ by $\nabla \ell(f_A, y)$, which is not a function of the residual alone, so both state and response must be observed; this is why the router head receives both. The bounds are conditional on the state and require the loss gradient at the current prediction; at routing time the label is unobserved and the router estimates the expectation of the first-order term from training episodes. The theory therefore explains the feature design and the second-order error of a response-based surrogate rather than providing a label-free guarantee.

B. From estimation error to routing regret

Theorem 5 (One-step routing regret). *Suppose that, at a fixed state A , the utility estimator satisfies $|\hat{m}(j | A) - m(j | A)| \leq \epsilon$ for every remaining candidate. If $\hat{j}$ maximizes the estimated utility and j^* maximizes the true utility, then*

$$m(j^* | A) - m(\hat{j} | A) \leq 2\epsilon. \quad (22)$$

Proof. By the error bound, $m(j^* | A) \leq \hat{m}(j^* | A) + \epsilon$. The maximization rule gives $\hat{m}(j^* | A) \leq \hat{m}(\hat{j} | A)$, and the error bound applies once more. $\square$

Combining Theorem 5 with Proposition 1 bounds the regret of a response-based router by twice the estimation error of its first-order surrogate plus the second-order response term. The theorem controls one routing step. A sequence of locally accurate choices can still visit a different state from exact greedy selection. When the induced set utility is normalized, monotone, and submodular, the standard approximate-greedy argument extends the local bound to

$$F(A_{\text{R-MUR}}) \geq \left(1 - \left(1 - \frac{1}{B}\right)^B\right) F(A^*) - 2B\epsilon, \quad (23)$$

where B is the context budget. The one-step result itself does not require monotonicity or submodularity.

VI. EXPERIMENTS

A. Experimental Setting

The controlled environment isolates the effect of redundancy and candidate-set size. It fixes a budget of four contexts and uses 16 candidates in the redundancy study; the candidate-scaling study varies the pool size over $K = 4, 8, 16, 32, 64$ at duplicate ratio .50. Candidate value is generated from a frozen subset-capable predictor, and Standalone Utility, the cached-only router, and R-MUR use matched encoder and head capacity. All controlled results are averaged over five seeds. The standalone oracle ranks candidates by the true standalone marginal and keeps that order fixed. The sequential oracle recomputes true marginals after every selection. The gap between these two policies defines structural headroom.

The real-data study uses six standard traffic benchmarks: METR-LA, PEMS-BAY, PEMS03, PEMS04, PEMS07, and PEMS08. The predictor is trained before routing and kept fixed; it accepts arbitrary candidate subsets. For each dataset, candidate pools are constructed from the training segment, the

TABLE I
CONTROLLED REDUNDANCY STUDY AT $K = 16$ AND BUDGET FOUR.
GAINS ARE LOSS REDUCTIONS RELATIVE TO THE ANCHOR-ONLY
PREDICTOR; RESULTS ARE AVERAGED OVER FIVE SEEDS.

Duplicate ratio	Standalone oracle	R-MUR	Sequential oracle	Structural headroom
0	.6880 $\pm$.0077	.6636 $\pm$.0101	.6880 $\pm$.0077	.0000
.25	.5161 $\pm$.0054	.5027 $\pm$.0105	.5813 $\pm$.0064	.1121
.50	.3512 $\pm$.0028	.3652 $\pm$.0030	.4471 $\pm$.0041	.2145
.75	.1771 $\pm$.0019	.2085 $\pm$.0049	.2481 $\pm$.0035	.2859

history length is 12, the forecast horizon is 12, the candidate count is 16, and the context budget is four. The primary comparison uses three seeds. The multi-target study fixes 32 evenly spaced target sensors per dataset and constructs a separate candidate pool for each target from the training split. We report the anchor-only predictor, pooling all candidates, relevance ranking, maximum marginal relevance, Standalone Utility, the cached-only router, R-MUR, the standalone oracle, and the sequential oracle. Unless stated otherwise, R-MUR uses a shortlist of four candidates and the ranking-calibrated objective of Eq. (13).

Two further protocols test the scope of the conclusions. The strong-backbone study replaces the ridge expert with a subset-capable spatio-temporal transformer, keeps every other component of the protocol identical, fixes 16 targets per dataset and uses three seeds; the shortlist size is swept over $q \in \{2, 4, 8, 16\}$ and the number of expert evaluations per decision is recorded. The second-family study re-instantiates the same router on demonstration selection for a frozen in-context classifier over five public image benchmarks, with the same candidate count, budget, shortlist sizes and response-feature semantics, replacing the forecast vector by the classifier’s class logits.

B. Structural Value of State-Conditioned Selection

In the controlled redundancy study, structural headroom grows monotonically with the duplicate ratio. Table I reports the oracle ladder. At zero redundancy the standalone and sequential oracles coincide. As redundancy increases, the gap reaches .286. At duplicate ratios .50 and .75, the learned router exceeds the standalone oracle even though that oracle uses the true standalone utilities. The gain therefore comes from re-evaluating candidate value after the selected state changes, not from a better standalone score.

The same structural effect appears on real benchmarks. Over six datasets, 32 fixed targets per dataset, and three seeds, every one of the 576 target-seed evaluations has positive structural headroom (minimum .0101). Table II summarizes the distribution over cells. The positive fraction is one for every dataset, and the mean headroom ranges from .076 to .119. State conditioning therefore creates measurable decision value across targets rather than at a single sensor.

C. Multi-Target R-MUR Benchmark

We next evaluate learned routing under the six-dataset protocol. The cached-only router first forms a shortlist of

TABLE II

REAL-DATA STRUCTURAL HEADROOM OVER 32 FIXED TARGETS PER DATASET AND THREE SEEDS (96 CELLS PER DATASET, 576 IN TOTAL). EVERY CELL IS POSITIVE; THE MINIMUM ACROSS ALL DATASETS IS .0101.

Dataset	Mean headroom	Median headroom	Positive fraction
METR-LA	.0959	.0844	1.000
PEMS-BAY	.1156	.1204	1.000
PEMS03	.1189	.1077	1.000
PEMS04	.0834	.0725	1.000
PEMS07	.0892	.0835	1.000
PEMS08	.0760	.0729	1.000

TABLE III

DEPLOYABLE SELECTION METHODS, INCLUDING THE TRIVIAL POOL-EVERYTHING POLICY, THE STRONGEST COMPETITOR ON FIVE OF THE SIX DATASETS. VALUES ARE PREDICTION GAINS AVERAGED OVER 32 FIXED TARGETS PER DATASET; THE FINAL COLUMN IS THE AVERAGE RANK ACROSS THE SIX DATASETS.

Method	METR-LA	PEMS-BAY	PEMS03	PEMS04	PEMS07	PEMS08	Avg. rank
Relevance (kNN)	-.0165	-.0199	-.0191	+.0071	-.0155	+.0074	5.67
MMR	-.0180	-.0249	-.0102	+.0098	-.0085	+.0075	5.33
Standalone Utility	+.0024	+.0272	+.0149	+.0248	+.0152	+.0246	3.67
Cached-only router	+.0042	+.0284	+.0154	+.0250	+.0150	+.0246	3.00
Pool all 16	+.0080	+.0254	+.0256	+.0365	+.0242	+.0359	2.00
R-MUR (ours)	+.0146	+.0557	+.0269	+.0357	+.0284	+.0333	1.33

four candidates, and R-MUR reranks that shortlist with compact predictor-response summaries. The response-aware model is trained with marginal-utility regression and the gap-weighted pairwise objective. Table III compares deployable selection policies, including the trivial pool-everything policy, which is the strongest competitor on five of the six datasets (on PEMS-BAY the cached-only router and the static-utility model edge past it). R-MUR has the best average rank (1.33 against 2.00), leads every specialised policy on all six benchmarks, and is *significantly* ahead of the best competitor on three of the six under target-level paired intervals (METR-LA $+.0067$ [$+.0041, +.0094$], PEMS-BAY $+.0303$ [$+.0203, +.0427$], PEMS07 $+.0041$ [$+.0015, +.0070$]). It is unresolved on two more (PEMS03 $+.0013$ [$-.0004, +.0030$], PEMS04 $-.0008$ [$-.0050, +.0025$]) and *significantly below* pooling on PEMS08 ($-.0026$ [$-.0042, -.0010$]). Relevance and maximum marginal relevance are harmful on four of the six datasets, showing that similarity and diversity do not reliably represent predictive utility; neither ever beats pooling.

Table IV reports the multi-target comparison. R-MUR improves over Standalone Utility and over the cached-only router on every dataset. Five datasets show positive target fractions of 94–100%. METR-LA shows the widest target-level variation, with 75% of targets positive and 81–84% exceeding the two deployable baselines. The response-aware stage uses predictor evaluations for only one quarter of the candidate pool.

The two contrasts test different components of the method. Improvement over Standalone Utility shows that state-conditioned marginal utility contains decision information absent from a fixed candidate score. Improvement over the cached-only router shows that predictor-response information and ranking calibration add value beyond a state-conditioned score built from cached representations alone.

Among the fourteen methods compared across the six

TABLE IV

MULTI-TARGET CONSISTENCY OF R-MUR. DIFFERENCES AND POSITIVE-TARGET FRACTIONS ARE COMPUTED OVER 32 FIXED TARGETS PER DATASET. ORACLE RECOVERY IS THE POOLED RATIO OF R-MUR GAIN TO SEQUENTIAL-ORACLE GAIN.

Dataset	Δ vs Static	Δ vs Cached	Positive targets	> Static	> Cached	Oracle recovery
METR-LA	.0123	.0104	.812	.969	.969	.099
PEMS-BAY	.0285	.0272	.938	1.000	.969	.309
PEMS03	.0121	.0115	1.000	1.000	1.000	.447
PEMS04	.0109	.0107	1.000	.969	.969	.534
PEMS07	.0132	.0134	1.000	1.000	1.000	.456
PEMS08	.0088	.0087	1.000	.969	.969	.516

benchmarks (the detailed two-dataset matrix below reports eleven of them) is a DELIFT-style rule (ICLR 2025) that combines submodular coverage with k-means representatives, implemented here as greedy facility location over an over-segmented pool of cluster medoids. It is significantly below pooling on all six datasets ($-.0196$ [$-.0241, -.0155$] on METR-LA, $-.0212$ [$-.0297, -.0128$] on PEMS-BAY, and between $-.018$ and $-.023$ elsewhere), so adding the most recent member of the submodular family does not change the verdict: the strongest competitor is still the trivial policy of using every candidate, and on every dataset the best non-trivial rule is greedy mutual information.

a) *Where a method change could still pay.*: Every remaining loss in this regime has one shape: the router wins on average while losing on a substantial minority of episodes, and on PEMS08 it loses on average. The per-episode gains all six deployments record put a ceiling on what any episode-level confidence gate could recover: taking the better of routing and pooling episode by episode would add $+.0148$ to $+.0778$ over pooling, three to seven times the router’s own advantage, and would convert the PEMS08 loss into a gain. The router loses on 39–52% of episodes, so the loss is a near-even split rather than a few outliers a safety rule could excise. We then tested whether that headroom is reachable, and it is not. Four signals available at decision time were recorded for every episode — the cached screen’s top-two margin, the utility model’s top-two margin, its top score, and the prediction-change norm of the candidate it picked — and each was scored for predicting whether the router beats pooling on that episode. All twelve dataset-by-signal AUCs lie in $[\.481, .521]$: the router’s own confidence carries no information about when it is wrong, and the best held-out gate adds at most $+.0030$ over pooling against a ceiling of $.013$ to $.048$. This is what Proposition 4 predicts rather than a defect of this router — the identifiable term is the anchor’s residual, a strong anchor has absorbed its systematic part, and what remains is noise, which no function of the same absorbed signal can anticipate. The remaining loss is therefore not a tuning, head-design or gating problem: the estimation error is present and unobservable.

D. *The standard selection toolkit is harmful when utility is estimable*

The ridge protocol is the regime in which context selection has realizable value, because the expert’s residual is systematic and therefore predictable. Table VII evaluates the full competitor set on the same 32 targets and three seeds. Two results stand out.

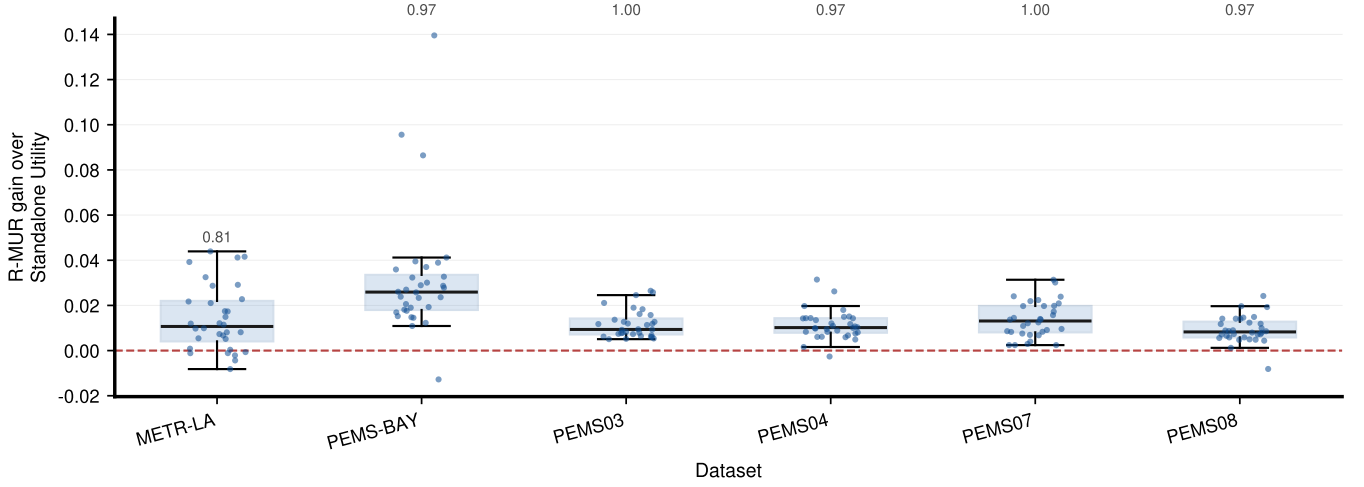


Fig. 2. Target-level difference between R-MUR and Standalone Utility over 32 fixed targets per dataset. Five datasets are positive on 94–100% of targets; METR-LA shows wider target variation.

First, *the default heuristics are significantly harmful on both datasets*: relevance ranking is .0165 below the anchor-only predictor on METR-LA and .0199 below on PEMS-BAY, MMR is .0180 and .0249 below, and on METR-LA DPP, k-center, facility location, k-means representatives, random subsets and even greedy mutual information are harmful as well, each with a target-level interval entirely below zero. Selecting correlated or diverse candidates by a proxy score injects noise that the predictor propagates; the oracles show that a great deal of value exists (+.133/+ .163 standalone), so this is a selection failure rather than an absence of signal.

The pattern generalises across the six benchmarks. Running the same training-free competitors on all six datasets with the frozen R080b cells for the learned policies (32 targets and three seeds each, 96 cells per dataset) gives one matrix in which the response-aware router is the best policy on five datasets and within rounding of the best on the sixth: METR-LA +.0146, PEMS-BAY +.0557, PEMS03 +.0269, PEMS04 +.0357, PEMS07 +.0284, PEMS08 +.0333, against pool-all +.0080, +.0254, +.0256, +.0365, +.0242, +.0359. Relevance ranking is *harmful* on four of the six datasets (METR-LA, PEMS-BAY, PEMS03, PEMS07) and maximum marginal relevance on four; on the two dense-flow datasets (PEMS04, PEMS08) the proxies are weakly positive, so the harmful regime is common but not universal.

Second, learning the decision recovers that value and more: R-MUR reaches +.0146, the only learned policy whose interval clears zero by a wide margin, roughly twice the strongest non-selection baseline (pooling all sixteen candidates, +.0080) and clearly above the response-free learned routers (+.0024 and +.0042, neither of which is significant on its own).

a) Why the default heuristic fails, and when it would not.: The sign of relevance ranking is not a property of relevance; it is a property of the candidate pool. Table VI holds the protocol fixed and moves the pool along the correlation spectrum. As candidates become less redundant with the anchor — mean $|\rho|$ falling from .77 to .43 — relevance ranking rises monotonically from harmful to helpful

TABLE V
OPERATING CHARACTERISTICS OF THE PROTOCOL USED AS A DECISION PROCEDURE. PILOTS OF n TARGETS ARE DRAWN FROM SIX RIDGE-REGIME DEPLOYMENTS (400 PILOTS PER DATASET AND SIZE); THE FULL-SAMPLE VERDICT IS OVER ALL 32 TARGETS. “AGREEMENT” IS THE STRICT THREE-WAY CALL.

Pilot size	Agreement	Sensitivity for route	Specificity for route	Pilot cost
4	.643	.664	.882	< 1 m
8	.769	.816	.940	< 1 m
16	.841	.859	.927	≈ 1.5 m
24	.959	.976	.957	≈ 2 m

(−.0086 → +.0006 → +.0063), and the whole pool becomes progressively less valuable (+.0116 → +.0070 → +.0012 for pooling everything) while the oracle falls only from +.150 to +.104. The practical rule is therefore conditional: retrieve the most similar context when candidates carry information the anchor does not already have, and do not when they are redundant with it. In the demonstration family the same heuristic is near-optimal because a candidate there contributes a *label*, which is exactly information the anchor lacks.

b) Turning the protocol into a deployment decision.: The measurement protocol is only useful if a practitioner can act on it before committing, so we measured its operating characteristics as a decision procedure. On the six ridge-regime deployments, pilots of n targets were drawn without replacement from the 32 available, 400 pilots per dataset and size, and each pilot was scored by the same rule applied to the full sample. Table V reports the result. A pilot of 24 targets reproduces the full-sample verdict 96% of the time, with 98% sensitivity and 96% specificity for the operationally critical “route” call; sixteen targets give 84%, 86% and 93%. These are not base-rate artefacts — half the deployments are genuinely positive, so a constant-“route” rule would score specificity zero — and the direction of the effect is far easier to recover than its significance: even four targets almost always get the sign right, while three-way agreement is only .64 there. In the ridge regime a cell costs two to five seconds, so the pilot is about

two minutes; with the nonlinear backbone it is about half an hour.

c) *Does the pilot survive a time shift?*: A pilot is necessarily labelled before the period the router will serve, so the procedure was re-run with an added evaluation split: the router and the pool-everything baseline are scored on the calibration period and on the test period for the same 32 targets of each dataset, 192 targets in all. Direction transfers well — the within-dataset correlation between the pilot and deployment differences is $+ .596$ ($p < 10^{-19}$), every dataset is individually significant ($+ .520$ to $+ .848$), and the per-target signs agree 81% of the time. The significance call transfers less well: the dataset-level verdict agrees on four of six, against the 96% of the same-period pilot. Both disagreements keep the correct sign, and, importantly, none of the four pilots that said “route” was contradicted by a deployment on which routing hurt. The honest reading is that the procedure is directionally robust to the time shift and its confidence is not, so the pilot should be enlarged rather than abandoned.

d) *Recognising a pilot that should not be trusted.*: The pilot can be audited against itself. Splitting it into two halves and computing the paired interval on each gives a check whose operating characteristics are measured on the same six deployments: at sixteen targets, pilots whose halves both resolve in the same direction agree with the full-sample verdict 99.0% of the time against 78.1% for the rest, and at twenty-four targets 99.9% against 92.3%. The check is high-precision and low-coverage — strict consistency holds for only 35% of sixteen-target pilots and 39% of twenty-four-target ones — so it certifies a verdict rather than producing one. Requiring merely that the halves share a sign is worth almost nothing (85.4% to 84.8% at sixteen targets). The refined procedure is therefore: split the pilot, act when both halves resolve in the same direction, and enlarge the pilot otherwise, rather than falling back on a cheaper proxy. One caveat is stated rather than hidden: requiring both halves to resolve selects for large effects, which are also the ones that resolve on the full sample, so the check is partly a signal-strength filter. A genuinely marginal deployment yields an inconclusive check, and the correct response there is more labels.

e) *Does the rule travel to another family?*: The demonstration family re-runs the same test in a different modality: its runs record the class of every evaluation episode, so half the classes can serve as the pilot and half as the deployment — a harder split than the traffic one, since classes are different sub-problems rather than draws from one population. Across four contrasts (router against pooling, relevance against pooling, MMR against pooling, and router against relevance) the pilot and deployment verdicts agree on all five benchmarks, and the per-target sign agreement is 1.000 throughout. The mechanism reproduces quantitatively as well: half-split verdict agreement collapses exactly where the effect is marginal relative to the pilot’s resolution — on CIFAR-10 the marginal contrasts are $\Delta \approx -.007$ over ten classes and two five-class halves agree only 31–34% of the time, while the same effect size over CIFAR-100’s hundred classes agrees every time.

One limitation is stated rather than glossed. Every contrast in this family has the same sign — the router and the response-

free rules are all below pooling — so 5/5 agreement is partly trivial and the test confirms specificity, not sensitivity. The demonstration family contains no positive case to call, because candidate demonstrations carry labels rather than redundant predictions. The operating characteristics of Table V therefore remain the traffic-family numbers, and the external evidence is about the mechanism, not the rates.

The procedure is therefore: label a pilot of roughly two dozen targets, run the protocol unchanged on it — screening, budget, objective, and both the router and the pool-everything baseline — compute the paired target-level interval for router minus pooling, and route only if it excludes zero above. Nothing cheaper substitutes. A fitted predictability probe does not work (Section VI-E), structural headroom does not work, and the cached screen is uncorrelated with the true marginal exactly in the regime where the decision is close.

The rule can be deployed without labels. *Redundancy-adaptive selection* computes one statistic — the mean absolute correlation between the anchor and the candidates, which the protocol already needs for relevance ranking — and then either keeps the top candidates by relevance (below a threshold) or includes the whole pool (above it). With the threshold chosen on half of the targets and then held fixed, the rule reaches $+ .0081$ [$+ .0017, + .0138$] on METR-LA and $+ .0129$ [$- .0066, + .0329$] on PEMS-BAY, against $- .0006$ and $- .0020$ for fixed relevance and $+ .0066$ and $+ .0098$ for always pooling everything; it captures 87% and 67% of an oracle that picks the better of the two policies per cell. On METR-LA the selected threshold is stable — $[.45, .55]$ in 198 of 200 splits — whereas on PEMS-BAY it concentrates at the low end of the grid instead (the minimum, $.40$, in 91 of 200 splits), so the threshold should be fitted per deployment rather than transferred. The evidence for the rule is resolved on METR-LA, where its interval excludes zero, and directional but unresolved on PEMS-BAY; on both datasets its advantage over always pooling is within noise ($+ .0015$ and $+ .0031$). What the rule establishes is that a label-free statistic can prevent a harmful default, not that it beats pooling. A training-free practitioner should therefore adapt the retrieval rule to pool geometry rather than apply it unconditionally; a practitioner who can learn the decision should use marginal-utility routing instead, which is roughly twice as good on the standard pools.

A single label-free statistic organises the whole table: *candidate novelty*, the share of a candidate’s variance that the anchor does not already explain (regress the candidate series on the anchor series on the training split and keep the residual share). Across the six pool geometries of the two datasets it correlates $+ .80$ with the relevance gain, and the two regimes separate without overlap: every harmful cell has novelty $\leq .46$ and every helpful cell has novelty $\geq .60$ (METR-LA $.40/.60/.79$, PEMS-BAY $.46/.73/.95$ for the correlated, moderate and independent pools). The statistic does *not*, however, transfer across datasets as a constant: over the six standard pools the correlation with the relevance gain is $-.33$, and the two helpful datasets sit at intermediate novelty ($.21-.24$) while two harmful ones sit above them (METR-LA $.40$, PEMS-BAY $.51$). Redundancy with the anchor is the mechanism that operates when the pool is manipulated inside

TABLE VI

THE SIGN OF RELEVANCE RANKING DEPENDS ON CANDIDATE-ANCHOR REDUNDANCY. RIDGE PROTOCOL, 8 TARGETS, THREE SEEDS, 72 CELLS.

Pool head	anchor $ \rho $	Relevance	MMR	Pool all	Oracle
most correlated	.77	−.0086	−.0092	+.0116	+.150
moderate	.63	+.0006	−.0062	+.0070	+.130
least correlated	.43	+.0063	+.0024	+.0012	+.104

TABLE VII

FULL COMPETITOR MATRIX IN THE RIDGE PROTOCOL (32 TARGETS, THREE SEEDS, 96 CELLS PER DATASET). GAINS ARE RELATIVE TO THE ANCHOR-ONLY PREDICTOR WITH TARGET-LEVEL 95% INTERVALS; A NEGATIVE ENTRY MEANS THE POLICY IS WORSE THAN USING NO AUXILIARY CONTEXT AT ALL.

Policy	METR-LA		PEMS-BAY	
	Gain	95% CI	Gain	95% CI
Sequential oracle	.1471	[.133, .161]	.1805	[.160, .201]
Standalone oracle	.1333	[.120, .146]	.1627	[.144, .181]
R-MUR $q=4$	.0146	[.009, .020]	.0557	[.044, .066]
Pool all 16	.0080	[.003, .013]	.0254	[.007, .044]
Cached-only router	.0042	[−.000, .009]	.0284	[.016, .041]
Standalone Utility	.0024	[−.002, .007]	.0272	[.013, .041]
Mutual information	−.0055	[−.010, −.001]	.0173	[.006, .028]
Random four	−.0077	[−.010, −.006]	.0095	[.003, .016]
k-means representatives	−.0080	[−.012, −.004]	.0043	[−.003, .011]
Facility location	−.0163	[−.023, −.009]	.0090	[−.005, .023]
Relevance (kNN)	−.0165	[−.024, −.009]	−.0199	[−.033, .004]
DPP	−.0166	[−.022, −.011]	−.0074	[−.023, .008]
MMR	−.0180	[−.025, −.011]	−.0249	[−.044, −.006]
k-center	−.0187	[−.025, −.013]	.0043	[−.003, .010]

one dataset; across datasets other factors dominate, which is why the protocol measures the statistic per deployment instead of transferring it. Redundancy with the anchor — not similarity as such — is what decides whether retrieval helps, which is also why the demonstration family behaves in the opposite way: there a candidate contributes a label the anchor lacks, so its novelty is high by construction even though its embedding is close to the query.

The sign flip replicates on the second dataset. Building the same three pool geometries on PEMS-BAY, relevance ranking moves from −.0333 (most correlated head, mean anchor $|\rho| = .65$) through +.0117 (.46) to +.0156 (.12), with MMR and DPP following the same ordering and the whole pool losing value (+.0112 → +.0064 → +.0119) while the oracle falls from +.179 to +.099.

E. Strong Nonlinear Backbone

The traffic results above use a subset-capable ridge expert, which is linear in the candidate histories. To test whether the conclusions depend on that function class we replace the expert with a subset-capable spatio-temporal transformer: per-node history embeddings, learned node embeddings, time-of-day and day-of-week embeddings, a three-layer token transformer over the anchor token and the selected candidate tokens with a key-padding mask, and an explicit linear path from the anchor history. The model is trained with subset dropout, so every subset size is in distribution; unselected candidates are provably inert (unit-tested). It is pretrained on the training

TABLE VIII

EXPERT STRENGTH ON IDENTICAL TRAINING EPISODES AND HELD-OUT WINDOWS (EIGHT TARGETS, SEED 101). RATIOS ARE THE ADAPTED NONLINEAR EXPERT OVER THE PER-TARGET RIDGE EXPERT.

Dataset	Mask family	Ridge	Nonlinear	Ratio
METR-LA	empty	.378	.364	.96
	four	.383	.360	.94
	full	.373	.353	.95
PEMS-BAY	empty	.550	.393	.72
	four	.549	.379	.69
	full	.487	.360	.74

split, then adapted on exactly the episodes the ridge expert sees, with the epoch chosen on the calibration split. The routing code path is unchanged: the same pair sampling, the same utility models, the same response features and the same ranking-calibrated reranker operate on the new expert through a common interface.

Table VIII verifies that the nonlinear expert is genuinely stronger than the ridge expert on the same episodes and the same held-out windows: after adaptation its mean squared error is lower in every mask family, by 4–6% on METR-LA and by 26–31% on PEMS-BAY. This is the intended stress test, because a better anchor-only predictor leaves less systematic residual for auxiliary contexts to correct.

Table XI reports the multi-target comparison on both datasets with 16 fixed targets and three seeds. Structural headroom is preserved: the sequential oracle gains .105 and .104 over the anchor-only predictor and the standalone oracle .090 and .088, so context selection still matters. The theory-shaped bilinear head of Section IV-A is the strongest selection policy in the matrix (.0107 at $q = 4$, .0112 at $q = 8$), ahead of the frozen head (.0097) and of every non-learned baseline, and it beats the cached-only router with a paired interval above zero (+.0031 [+ .0009, +.0057]) while coming within a hair of beating Standalone Utility (+.0037 [−.0001, +.0089]). It does not, however, beat the trivial policy of pooling all sixteen candidates (.0133). *The original gate is therefore not passed on either dataset* — the frozen head’s contrasts are +.0028 [−.0012, +.0083] and +.0021 [−.0006, +.0053] on METR-LA and −.0008 [−.0036, +.0019], −.0003 [−.0024, +.0019] on PEMS-BAY — whereas the same contrast is clearly positive with the ridge expert. On PEMS-BAY every learned router, including R-MUR, is also below plain random selection (.0161).

The two datasets resolve the comparison against pooling differently, and paired tests are needed to say which. On PEMS-BAY pooling is significantly above every alternative, response-free or response-aware: the paired difference is −.0094 [−.0135, −.0053] for the frozen response-aware head and −.0091 [−.0126, −.0057] for the cached-only router, and it is also above random (−.0063 [−.0079, −.0048]). On METR-LA pooling is significantly above every response-free rule — relevance −.0062 [−.0093, −.0029], MMR −.0057 [−.0089, −.0022], Standalone Utility −.0064 [−.0115, −.0009] — but *not* above

the response-aware family ($-.0036 [-.0110, +.0067]$ at $q = 4$, $-.0025 [-.0093, +.0075]$ at $q = 8$) and not above greedy mutual information. The response signal is thus the only thing that keeps a learned router in contact with pooling once the predictor is strong; whether it closes the gap is dataset-dependent and unresolved here. Both remain an order of magnitude below the oracle (.090–.105), so the failure is one of realisability rather than of ranking.

The compute profile explains part of the difference. Each decision spends $B(1 + q)$ expert evaluations: 20 per episode at $q = 4$ and 36 at $q = 8$, against 64 for reranking the full pool and 17 for the standalone oracle. Measured at deployment batch size on the same cell, the response-aware stage costs .74, 1.19, 1.59 and 2.65 milliseconds per episode for $q = 2, 4, 8, 16$, against 1.70 milliseconds for the sequential oracle and .03 milliseconds for cached screening alone, and it raises peak device memory from 90 to 173 megabytes. The shortlist therefore retains its intended efficiency, and the trend across shortlist sizes ($q = 4$: .0097, $q = 8$: .0108) shows that the response stage keeps improving as it sees more candidates, i.e. the cached screen, not the response stage, is the limiting component with a strong expert.

Among training-free policies, one stands out in this regime: greedy mutual information — choose the candidates that most reduce the conditional variance of the target under the pooled training covariance — reaches .0134, a statistical tie with pooling everything (.0133, paired $+.0001 [-.0040, +.0043]$) and nominally above every learned router, including ours (frozen $q=8$: paired $+.0026$, n.s.; bilinear $q=8$: $+.0022$, n.s.). The other geometry policies are far below (k-center .0060, k-means representatives .0076, consensus alignment $-.0046$). When utility cannot be estimated, a label-free information criterion is therefore the best *selection* rule — again consistent with the identifiability reading, and a reminder that in this regime the choice of objective matters less than whether the objective is estimable at all.

A second nonlinear backbone, a permutation-invariant masked-mean network, reproduces the ordering but not the significance: over eight targets and two seeds pooling every candidate remains first (.0295), the response-aware router at $q = 8$ is statistically tied for second (.0282, paired $-.0013 [-.0034, +.0007]$ against pooling) while at $q = 4$ it is significantly below (.0265, $-.0030 [-.0059, -.0004]$), and determinantal diversity (.0279), relevance (.0259), the cached router (.0258), Standalone Utility (.0250) and random selection (.0242) follow, with ΔStatic $+.0015 [-.0010, +.0046]$ against an oracle gap of .0194. Across both backbones the pattern is stable: the learned routers capture roughly a quarter of the standalone-oracle gain on this reduced target set, the ordering favours R-MUR, and the paired advantage over the response-free routers is smaller than the target-level noise at these sample sizes.

a) *The budget itself is the wrong lever.*: Figure 3 (left) shows why neither better heuristics nor budget tuning closes the gap. With the true marginals the best budget is *three* contexts (+.105), and every additional context reduces the gain: the oracle is not monotone in the budget. Without identifiability the picture inverts — random k -subsets improve

TABLE IX
POOL-SIZE SCALING WITH THE STRONG BACKBONE (METR-LA, 16
MATCHED CELLS, BUDGET FOUR). ONLY THE CANDIDATE COUNT
CHANGES BETWEEN ROWS.

Candidates	Pool all	R-MUR	Random four	Oracle
16	+.0057	+.0076	+.0005	+.0851
32	+.0128	+.0031	+.0049	+.0987
64	+.0081	-.0033	+.0018	+.1180
128	+.0042	-.0025	-.0006	+.1196

monotonically with k — so the best deployable policy in the entire matrix is to include all sixteen candidates (+.026). The two curves move in opposite directions through the protocol’s budget of four, and the vertical distance between them at $k = 4$ is the identifiability gap. Figure 3 (right) states the same gap as a realised share: every deployable policy, including all learned routers and the policy-gradient selector, recovers under 15% of the standalone oracle, and the trivial pool-everything policy recovers the most.

b) *Neither is the size of the library.*: The most natural escape from this boundary is that the protocol’s pool is too small: with a realistic library of hundreds of candidates, pooling all of them should be infeasible and selection should win. Table IX tests that directly by holding the budget at four and growing the pool from 16 to 128 candidates on the same sixteen cells.

The premise of the objection is correct and the conclusion is the opposite of what it suggests. The opportunity grows as claimed — the standalone oracle rises from +.0851 to +.1196, a 40% increase, and the paired growth of the identifiability gap is resolved at every step (+.0181 [+0.0126, +.0239] from 16 to 32 candidates, +.0257 [+0.0197, +.0315] from 32 to 64, and +.0446 [+0.0376, +.0514] over the full range, saturating between 64 and 128). But R-MUR captures *less*, falling from +.0076 to $-.0025$. At 32 and 64 candidates it is significantly *below* pooling ($-.0097 [-.0159, -.0022]$ and $-.0114 [-.0192, -.0018]$), and beyond 16 candidates it is indistinguishable from selecting four at random (paired $-.0018$, $-.0052$ and $-.0019$ at 32, 64 and 128). Enlarging the shortlist does not repair it — $q = 8$ minus $q = 4$ is +.0011 at 16 candidates and $-.0014$, $-.0016$ at 64 and 128 — which rules out the simple fix of buying more shortlist slots, though it does not by itself separate a noisy screen from an unidentifiable marginal. Pooling, meanwhile, is flat in pool size, so R-MUR’s decline is not caused by pooling degrading.

The binding constraint is identifiability, not the supply of candidates. Retrieving from a larger library raises the value that exists and lowers the share of it that can be identified, so it makes the routing problem harder rather than easier — which is the opposite of the usual scaling intuition and the reason we state the boundary rather than hide it.

One alternative explanation remains, and we removed it directly. If the failure came from the cached screen — which does correlate with the true marginal at chance level in this regime — then giving the response head an unrestricted view of the pool should repair it. Running the same cells with $q = K = 16$, so that no candidate is discarded before the

TABLE X

TEST-TIME ADAPTATION SWEEP ON METR-LA (16 MATCHED CELLS).
ONLY THE NUMBER OF FINE-TUNING EPOCHS CHANGES BETWEEN ROWS.
SHARE IS THE LEARNED GAIN DIVIDED BY THE STANDALONE ORACLE.

Epochs	Anchor MSE	Learned gain	Pool all	Oracle	Realized share
0	.4364	+.0169	+.0124	+.0858	.104
2	.4056	+.0035	+.0068	+.0831	.057
5	.4023	+.0070	+.0046	+.0842	.058
10	.4003	+.0076	+.0057	+.0851	
30	.4006	+.0078	+.0059	+.0853	

head sees it, changes the result by $+.0012$ $[-.0023, +.0044]$: the router reaches $+.0088$ against $+.0076$ for $q = 4$, still not above pooling ($+.0029$ $[-.0077, +.0193]$) and still not above random four ($+.0082$ $[-.0029, +.0243]$). The screen is therefore a *symptom* of the identification failure rather than its cause, and no improvement to screening can recover the value. It also means the two-stage design is justified on its own terms: at this pool size the shortlist is free in value terms while costing 20 expert rows per episode instead of 64.

A third test asked whether selection becomes valuable when candidates can hurt. We rebuilt the pools from the *least* correlated sensors, so the head of the pool is actively uninformative, and repeated the comparison on eight targets and two seeds. Pooling everything has the highest point estimate ($+.0040$) and R-MUR the lowest ($+.0013$), against oracle gains of $+.066$ and $+.082$: an expert trained with subset dropout is robust to irrelevant contexts, which removes the leverage selection would need. The ordering within the regime should not be read as a ranking, because none of it is resolved: paired against pooling, R-MUR is $-.0027$ $[-.0072, +.0020]$, relevance $-.0005$ $[-.0033, +.0023]$ and random $-.0009$ $[-.0029, +.0013]$, and only facility location is significantly below. The pool construction matters more than the policy here: on a separately built anticorrelated pool (24 cells, three seeds) relevance is significantly *above* pooling ($+.0051$ $[+.0025, +.0077]$). What is stable in this regime is the size of the remaining gap to the oracle, not which policy is nominally ahead. With a strong predictor, what the router feeds the expert barely matters for accuracy — and the value that remains is exactly the part that cannot be identified.

c) Adaptation is the operation that destroys the value.:

The ladder varied the architecture, so strength and inductive bias moved together. Per-target fine-tuning isolates the mechanism with a one-parameter sweep: same backbone, same pools, same episodes, only the number of adaptation epochs changes. Table X reports five levels on 16 matched cells (eight targets, two seeds).

Three things are established and one tempting claim is refused. First, the structural opportunity is *invariant*: the standalone oracle moves only between .0831 and .0858 across the whole range, so selection matters exactly as much after thirty epochs of adaptation as before any. Second, the loss from adapting is large and resolved: paired across cells, the realized share falls $-.140$ $[-.197, -.085]$ and the learned gain $-.0134$ $[-.0198, -.0075]$ between zero and two epochs, and the endpoint contrast is also resolved ($-.106$ $[-.170, -.030]$).

Two epochs — a 7.1% MSE improvement — remove roughly five sixths of the value routing can realize. Third, the accuracy that later epochs buy is negligible ($.4056 \rightarrow .4006$ over the next twenty-eight epochs), so the identifiability step coincides with the first adaptation step, where the residual turns from mostly systematic into mostly noise, and not with any later accuracy gain.

The same sweep also identifies the one setting in this paper where the strong backbone produces a *resolved* component for the response signal. Against the Standalone Utility ablation, R-MUR gains $+.0122$ $[+.0061, +.0198]$ with an unadapted backbone but only $+.0024$ $[-.0032, +.0092]$ after two epochs and $+.0027$ $[-.0030, +.0098]$ after thirty. The response-aware head is therefore not merely inert under a strong predictor: it adds resolved value exactly while a systematic residual remains, and that window closes as soon as the predictor is adapted to the deployment. What we do *not* claim is the partial recovery visible in the share column: its paired increments are $+.027$ $[-.036, +.119]$, $+.006$ $[-.002, +.016]$ and $+.0015$ $[+.0000, +.0046]$, and the two-to-thirty-epoch contrast is $+.035$ $[-.019, +.122]$. The defensible description is a step down onto a flat plateau.

The direction of the greedy search is not the problem. Running selection backwards — from the full pool by elimination — matches forward greedy to within $+.0002$ ($[-.00003, +.0005]$, backward better on 18 of 24 cells), so the utility landscape behaves submodularly for search purposes. What limits performance is what can be identified before the outcome is observed, not how the search proceeds.

Target-level structure reinforces that reading. Over the sixteen targets of the strong-backbone protocol, R-MUR recovers 10.7% of the standalone-oracle gain on average against 7.6% for Standalone Utility, and its per-target gain tracks the available oracle gain closely (Pearson $+.912$, $p < 10^{-4}$), so it does capture more where there is more to capture. What it does *not* do is track structural headroom: the correlation between a target's H_{state} and R-MUR's advantage over Standalone Utility is $+.047$ ($p = .86$). Headroom is necessary for sequential routing to matter, but with a strong backbone it does not predict where a router will succeed, because the binding constraint is what the router can observe rather than what the oracle could exploit.

The same test in the regime where routing *does* work makes the negative general. Over the six-dataset ridge protocol, 32 targets each (192 targets in total), the within-dataset correlation between a target's structural headroom and the router's advantage over pooling is $+.040$ ($p = .58$); per dataset it ranges from $-.235$ to $+.310$ with no consistent sign. Across datasets the mean headroom and the mean advantage are more alike ($r = +.60$, $p = .21$, $n = 6$) but the ordering is broken by the most informative case: PEMS03 has the *highest* mean headroom of the six (.119) and an advantage of $+.0013$, while PEMS-BAY has similar headroom (.115) and the largest advantage ($+.0303$). Structural headroom is therefore a necessary condition — a router cannot capture what the oracle cannot reach — and not a usable predictor of routing benefit in either regime. What does discriminate is the *realised* share measured under the deployment's own protocol,

TABLE XI
STRONG-BACKBONE COMPARISON (16 TARGETS, THREE SEEDS, $K = 16$,
BUDGET FOUR, SHORTLIST SIZE q). GAINS ARE LOSS REDUCTIONS
RELATIVE TO THE ANCHOR-ONLY PREDICTOR; Δ ROWS ARE
TARGET-LEVEL BOOTSTRAP INTERVALS AGAINST THE TWO
RESPONSE-FREE LEARNED ROUTERS.

Policy	METR-LA	PEMS-BAY
Anchor only	.0000	.0000
Random four	.0061	.0161
Relevance	.0070	.0107
MMR	.0076	.0113
Diversity (DPP)	.0091	.0142
Pool all candidates	.0133	.0224
Standalone Utility	.0068	.0138
Cached-only router	.0076	.0133
Learned set selector	−.0041	—
R-MUR ($q = 4$)	.0097	.0130
R-MUR ($q = 8$)	.0108	.0141
Standalone oracle	.0904	.0880
Sequential oracle	.1051	.1037
Δ Static, $q = 4$	[−.0012, +.0083]	[−.0036, +.0019]
Δ Cached, $q = 4$	[−.0006, +.0053]	[−.0024, +.0019]
Targets positive	14/16	15/16
Targets above Static	9/16	5/16
Δ Static, learned selector	[−.0177, −.0042]	—

TABLE XII
FULL COMPARISON ON THE STRONG BACKBONE (METR-LA, 16
TARGETS, THREE SEEDS, 48 CELLS). Δ IS AGAINST POOLING ALL
CANDIDATES, THE STRONGEST DEPLOYABLE ROW.

Policy	Gain	Δ vs pool-all
Sequential oracle	.1052	+ .0918
Standalone oracle	.0904	+ .0771
Pool all 16 (no selection)	.0133	—
R-MUR bilinear $q=8$ (ours)	.0112	−.0021
R-MUR bilinear $q=4$ (ours)	.0107	−.0027
R-MUR frozen $q=8$	.0108	−.0025
R-MUR frozen $q=4$	.0097	−.0036
Diversity (DPP)	.0091	−.0042
MMR	.0077	−.0057
Cached-only router	.0076	−.0058
Relevance	.0071	−.0062
Standalone Utility	.0069	−.0064
Random four	.0062	−.0071
Facility location	.0041	−.0092
Minimum disturbance, greedy (theory limit)	.0022	−.0112
Anchor only	.0000	−.0133
Policy-gradient selector	−.0041	−.0174

and that is what the diagnostic protocol of Section VI-E should be read as measuring.

A corollary worth stating, since it bears on how the six-dataset result should be read: the datasets differ by effect size rather than by statistical power. The per-target standard error of the router-versus-pooling difference is comparable across benchmarks (.0008 to .0059), while the effect itself ranges from $t = +5.1$ on PEMS-BAY (94% of targets favour routing) through $t = +1.5$ on PEMS03 (69%) to $t = -3.1$ on PEMS08 (38%). PEMS03 is a genuinely small effect, not an under-powered one: resolving it at its observed size would need roughly three times as many targets.

The theory’s own limit case was tested next, and it fails. As the anchor absorbs its residual the coefficient g_A of

Equation (21) goes to zero, so the score collapses to the observable curvature $-\|d_{A,j}\|^2$ and the optimal action should be to select the candidates that perturb the predictor *least*. That prediction is sharp and falsifiable, and it is refuted: the least-perturbing policy gains +.0026 on METR-LA and +.0037 on PEMS-BAY against pooling’s +.0133 and +.0224, i.e. it is significantly *below* pooling on both (−.0106 [−.0163, −.0051] and −.0188 [−.0227, −.0150]) and ranks thirteenth and sixteenth of the eighteen non-oracle policies. The sign of the theory’s limit is visible — it is the best of the six response-geometry rules on METR-LA — but a score that only avoids harm cannot create value: the candidates that barely move the predictor are also the ones that contribute least. The regime-B boundary therefore lies in the absence of the interaction term $\langle g_A, d_{A,j} \rangle$, not in how the score is written.

Two features of this table are worth stating plainly. First, the learned response-free routers are statistically indistinguishable from random selection (.0068 and .0076 against .0061), while the oracles show that a large amount of utility is available: with a strong expert, marginal utility is present but not recoverable from the router’s cached view. A learned policy-gradient selector that samples subsets and trains directly on their realised utility, with no marginal-utility model at all, does worse still (−.0041, i.e. .0109 below Standalone Utility with a paired interval [−.0177, −.0042]), which locates the failure in the observability of the decision rather than in the choice of utility estimator. Second, pooling all sixteen candidates is competitive with every selection policy, which means the strong expert tolerates irrelevant contexts and the value of selection lies in finding the few candidates that help rather than in avoiding harm. Figure 4 (left) shows the same comparison graphically against the two oracle levels.

Because a cheap probe would be valuable as a deployment gate, we tested that use directly and report that it fails. On the two extremes of the expert ladder of Table XIII — the one-layer transformer and the three-layer one — with three targets and 2,000 training and 1,200 test episodes each, the probe’s out-of-sample R^2 stays within noise of zero for both (−.004 ± .019 and −.015 ± .028), and its value statistic must be read against a per-cell random baseline that ranges from .115 to .294 of the oracle; excess over random is +.048 and +.067 of the oracle with a standard deviation of .064 across the six cells, while the ladder effect on the same two rungs is .205. A statistic whose excess is a quarter of the effect it is meant to explain, and whose noise exceeds that excess, cannot gate deployment: any threshold low enough to admit the strong expert also admits the weak one. We therefore do not propose a probe-based fall-back rule. The probe’s role in this paper is diagnostic — it establishes *why* the ordering is missing — while the executable test of identifiability remains the routing protocol itself, run with the screening, budget and objective of the deployment.

A closed-form probe makes the observability limit precise. On one cell we assembled the full set of quantities a router can see at decision time — anchor history, candidate history, candidate identity, the response vector, its magnitude and the observable base-response interaction — and fitted ridge and small multi-layer heads to the true standalone marginals of

TABLE XIII
EXPERT LADDER ON METR-LA (EIGHT TARGETS, SEED 101). A STRONGER EXPERT HAS A LOWER ANCHOR-ONLY MSE.

Expert	Anchor MSE	Standalone oracle
ridge (linear, out of family)	.3782	+.1535
transformer, 1 layer $d=32$	.3721	+.0835
masked-mean MLP	.3625	+.1002
transformer, 2 layers $d=64$	.3578	+.0923
transformer, 3 layers $d=128$	.3562	+.0902

the adapted expert. The oracle gain of the best standalone four is $+.245$ and the expected gain of a random four is $+.031$, so selection matters. Yet every probe reaches a *negative* out-of-sample R^2 ($-.03$ for state features, $-.20$ for state plus response) and a mean within-episode rank correlation with the true marginal of at most $.04$ in magnitude. The observable state carries the scale of the utility but not the ordering of candidates within an episode, which is the quantity a router needs; this is the regime predicted by the decomposition of Equation (21), where the identifiable term is the residual of a predictor that has already absorbed the systematic part.

d) The theory's central prediction, tested directly.:

Proposition 4 implies that the identifiable term is the fixed predictor's own residual, so a *stronger* predictor should leave *less* realizable value while the structural opportunity persists. We built a ladder of experts on identical cells (eight targets, seed 101, same pools and episodes): three spatio-temporal transformers of increasing capacity, a masked-mean MLP, and the per-target ridge expert of the frozen protocol as an out-of-family contrast. Strength is the anchor-only mean squared error on held-out windows; realizable value is the learned router's share of the standalone-oracle gain.

Within the neural family the prediction holds as a monotone dose-response: the structural opportunity is flat ($+.084$ to $+.100$ standalone, $+.096$ to $+.120$ sequential) while the realized share falls by a factor of 4.3 and the learned gain by a factor of 3.0 as anchor-only error falls from $.372$ to $.356$. Across the four neural rungs anchor-only error orders realized share perfectly (Spearman $+1.0$), with a fitted slope of 10.5 realized-share points per unit anchor MSE: a 1% anchor improvement corresponds to about $+.038$ share. A 4% improvement in the expert removes three quarters of the value that routing can realise.

The ridge rung is the informative exception, and it sharpens the theory rather than contradicting it. The ridge expert has the *worst* anchor-only error ($.3782$) yet the *largest* structural opportunity ($+.1535$, because its residual is highly systematic) and the largest absolute learned gain ($+.0313$); its realized *share* ($.198$) is nonetheless below the weakest transformer ($.268$), though above the two larger ones ($.177$ and $.063$), because its denominator is larger. Including it, the rank correlation between anchor error and realized share falls and the slope halves to 4.9. Residual magnitude is a sufficient statistic for realizable value only *within* a fixed hypothesis class; across families the inductive bias changes which candidates are jointly improvable, so accuracy and residual geometry decouple. Anchor-only accuracy is therefore not a sufficient

TABLE XIV
SECOND TASK FAMILY. MEAN CROSS-ENTROPY REDUCTION (NATS) AT $q = 4$; Δ COLUMNS ARE PAIRED CONTRASTS AGAINST THE TWO RESPONSE-FREE LEARNED ROUTERS. "H-STATE" IS THE SEQUENTIAL-ORACLE-STANDALONE ORACLE GAP.

Sequential oracle	Learned gain	Standalone oracle	Realized share			
Benchmark	$+.1687$ $+.0961$	R-MUR	$+.0313$ $+.0222$	Static	Cached	$+.198$ $+.268$
CIFAR-10	.0643	.0279	.0482	.0434	+.0160	.000
CIFAR-100	.5247	.0165	.5769	.5381	$-.0522$	.000
SVHN	.8809	.0075	.5530	.7688	+.1279	.000
EuroSAT	.1294	.1074	.1125	+.0221	.000	.000
DTD	.3322	.5981	.6478	$-.02658$	.000	.000

statistic for identifiability across architecture families: what matters is how much of the residual stays predictable, and that quantity has to be measured, not inferred from a leaderboard.

F. Second Task Family: Demonstration Selection

To test whether the same decision object and the same method transfer outside forecasting, we instantiate R-MUR on context-example selection for a frozen in-context image classifier. Candidate contexts are labelled training images; the anchor is a query image; the fixed predictor is a transformer that reads the query embedding and the embeddings and labels of the selected demonstrations; utility is the reduction of query-batch cross-entropy. Embeddings come from a frozen CLIP ViT-L/14 encoder. Five public benchmarks are used (CIFAR-10, CIFAR-100, SVHN, EuroSAT, DTD) with three seeds, $K = 16$ candidates, budget four and 1,602 test episodes per benchmark. The router is re-instantiated unchanged, including the seven-dimensional response summary, now computed in class-logic space.

The predictor is genuinely in-context: mean accuracy rises from 0.966 to 0.983 on CIFAR-10, 0.812 to 0.997 on CIFAR-100, 0.693 to 0.996 on SVHN, 0.946 to 0.986 on EuroSAT and 0.728 to 0.921 on DTD when four relevant demonstrations are added, and the farthest demonstrations are worse than random ones. Table XIV reports the gate. R-MUR beats both response-free learned routers with a paired confidence interval above zero on three of five benchmarks, and fails on CIFAR-100 and DTD, where a plain nearest-neighbour relevance rule is already within a fraction of a percent of the oracle.

The second family also exposes a boundary condition of the framework. The sequential and standalone oracles coincide to four decimals on all five benchmarks ($H_{\text{state}} \leq 5 \times 10^{-4}$): because the candidate pools contain no redundant demonstrations, selecting the best standalone demonstrations is already optimal, and state-conditioning has nothing to exploit. A nearest-neighbour relevance rule attains 99.4–99.9% of the oracle gain on every benchmark. The family therefore tests transfer of the ranking machinery, not the state-conditioning mechanism that drives the forecasting results, and it identifies the regime in which a simple baseline is hard to beat: no redundancy among candidates and a predictor that is already near-saturated with a handful of contexts. Figure 4 (right) shows the paired contrasts per benchmark.

A redundancy-structured variant tests the mechanism directly. We rebuilt the candidate pools so that each con-

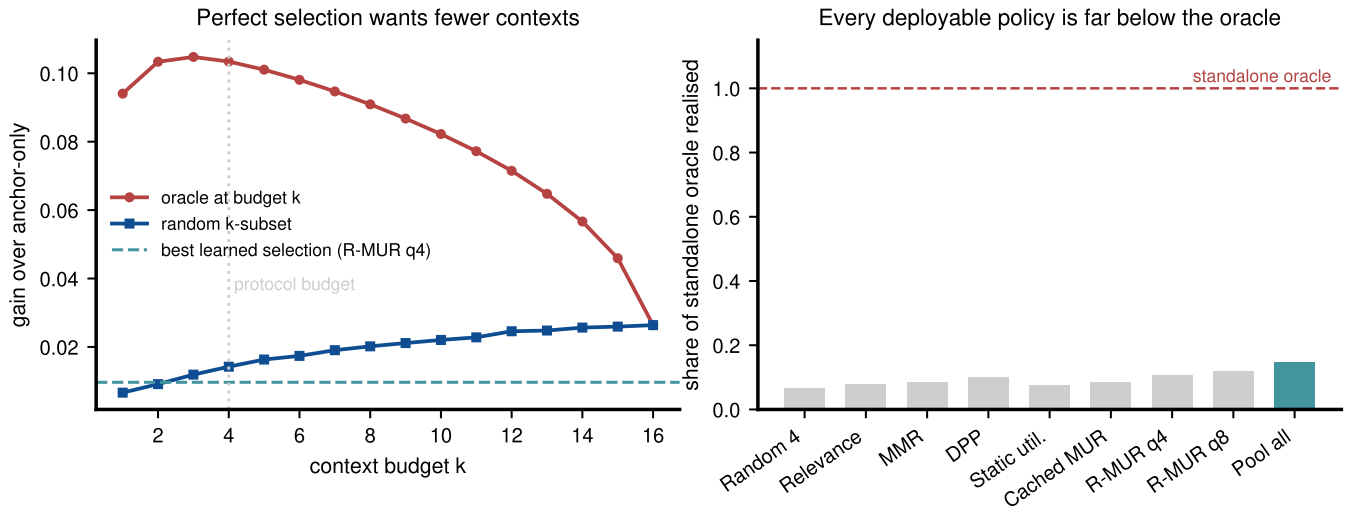


Fig. 3. The identifiability gap under a strong predictor. Left: with true marginals the optimal budget is three contexts and the oracle decreases with every additional context, while random subsets improve monotonically, so the best deployable policy in the matrix is to include all sixteen candidates; the best learned selection is below random four. Right: share of the standalone-oracle gain realised by each deployable policy.

tains four visual clusters with three near-duplicate members each (measured within-cluster cosine similarity .999 for controlled copies against .575–.838 between clusters), and pre-registered two predictions: redundant pools should create state-conditioning headroom, and R-MUR should then beat both response-free learned routers. The first prediction fails in substance. Redundancy is real — a second demonstration from the same cluster retains only 6.7–42% of the first member’s marginal — yet the sequential oracle exceeds the standalone oracle by at most 0.09% of its gain. The measured reason is that demonstration utility is label-driven: replacing candidate features by noise costs only 18% of the first-member marginal and leaves later marginals unchanged, whereas equalising labels flips every marginal negative. For this predictor the utility is close to a function of the label multiset of the selected set, so no pool geometry can make it state-conditioned.

We also iterated on the scoring head itself, in three steps, to test whether the failure is one of functional form. Replacing the scalar head by the theory-shaped bilinear score of Equation (16) improves the ranking of the true marginal on four of five benchmarks (mean within-state Spearman .375 against .156; top-one agreement .67 against .08 on CIFAR-100) and gives the largest pooled forced-budget contrast against Standalone Utility of the three heads, +.0605 [−.0036, +.1343], with three of five per-benchmark intervals above zero. The residual failure has a precise cause: the association between response magnitude and marginal utility changes sign across benchmarks (+.257 on CIFAR-10, −.590 on CIFAR-100), while the identity’s curvature term has a fixed sign. Releasing that constraint — a learned, signed, unconstrained curvature coefficient — confirms the diagnosis directly: the coefficient moves from its −.25 initialisation to +.14, +.13, +.11 and +.08 on four of the five benchmarks (significantly different from both −.25 and 0, $p < 10^{-6}$), and is ≈ 0 only on SVHN, exactly where the correlation with the curvature term was weakest. But the gain does not follow. The signed head’s

pooled contrast is +.0528 [−.0081, +.1236], it passes the per-benchmark criterion on two of five rather than three, and it is uniformly no better than the fixed-sign head; the extra unconstrained unit re-fits the same ordering rather than improving it, because the learned first-order coefficient already absorbs the ordering signal the curvature carries. Four head variants — scalar, curvature-corrected, bilinear and signed bilinear — therefore bracket the same result, which locates the ceiling in what the router can observe rather than in how the score is parameterised.

The second prediction holds once the stopping rule is separated from the ranking. The frozen rule stops when the predicted marginal becomes non-positive, which truncates selection to 0.53–0.63 of the budget on DTD and fails the gate (pooled $\Delta_{\text{Static}} - .162$). Forcing the budget to be spent passes on five of five benchmarks in both pool environments, with pooled $\Delta_{\text{Static}} +.045 [+ .018, +.080]$ and $+.055 [+ .020, +.101]$ and $\Delta_{\text{Cached}} +.070$ and $+.108$, all with paired confidence intervals above zero and passing under class-clustered intervals as well. The transferable part of R-MUR is therefore its response-aware ranking; the threshold is a calibration-dependent stopping device that should be fitted per family rather than inherited.

G. What Determines Learned Routing Quality?

Candidate-pool growth compresses the decision margin. In the controlled scaling study, absolute utility error improves as K increases, but the standard deviation of true marginals falls faster. Normalized error rises from .403 to .739, the positive top-two margin falls from .220 to .024, and the median ratio of maximum candidate error to that margin rises from 1.17 to 3.40. Decision accuracy falls from .884 to .147, while one-step regret rises from .018 to .067 at $K = 16$ and remains elevated at .049 at $K = 64$. Large candidate pools therefore make the argmax decision harder even when pointwise regression error improves in absolute units.

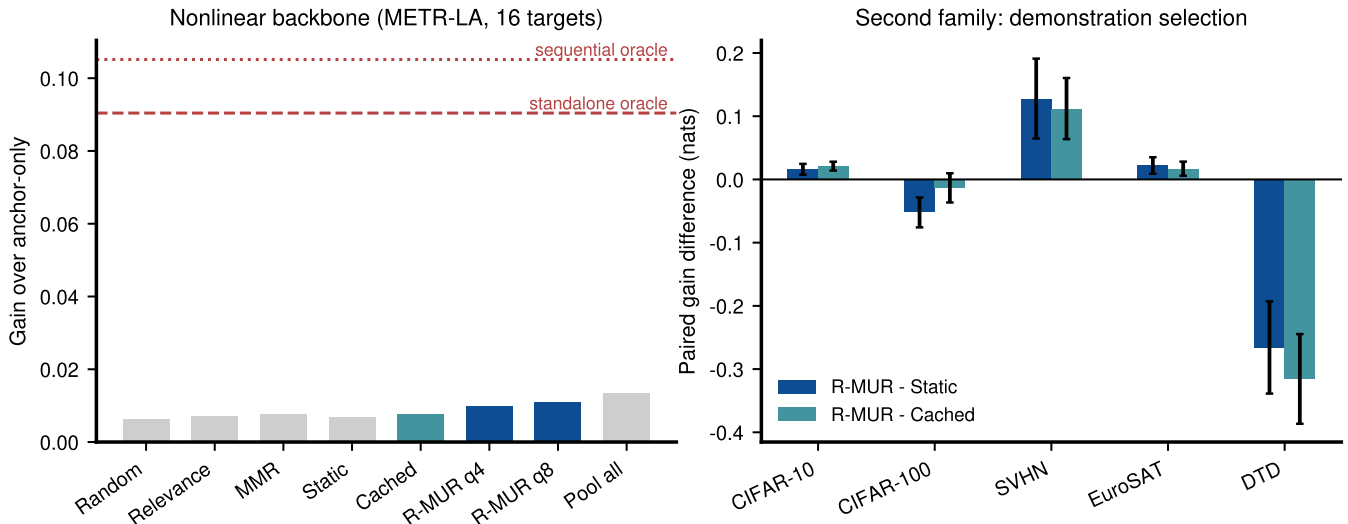


Fig. 4. Generality evidence. Left: with the subset-capable nonlinear backbone on METR-LA, every deployable policy remains far below the two oracle levels, and the learned response-free routers are indistinguishable from random selection. Right: in the second task family, R-MUR improves on both response-free learned routers on three of five benchmarks and degrades on the two where nearest-neighbour relevance is already within a fraction of a percent of the oracle. Bars are paired 95% intervals.

Predictor response determines whether marginal utility is observable. With cached context features, the response-feature comparison reaches a Spearman correlation of .164 and positive-utility AUC of .649. Adding compact response summaries raises them to .391 and .834, improves top-1 accuracy from .148 to .260, and reduces one-step regret from .0736 to .0569. Full forecast trajectories add little beyond the compact summaries. These results motivate the response-aware stage of R-MUR.

Finally, a calibration-derived stopping threshold traces a gain-harm frontier. At harmful fraction .25 the uncalibrated router reaches a gain of .273 with a negative-transfer rate of .032; calibration moves the same policy to a gain of .072 with harm .0001, i.e. it buys an elevenfold reduction in harm for three quarters of the gain. The same trade appears at higher harmful fractions: at .50 the uncalibrated harm is .085 and at .75 it is .227, against calibrated harm of .0006 and .0012. The threshold changes only the stopping decision; the candidate ranking remains the response-aware score. Across all settings, R-MUR is greedy and targets one-step decision quality under a context budget.

VII. DISCUSSION

The experiments support a two-stage design test for context-routing systems. The first stage measures structural headroom: whether state conditioning creates value that a perfect standalone ranking cannot reach. The second stage measures how accurately marginal utility can be estimated from information available to the router. Tasks with little headroom need no sequential marginal routing. Tasks with substantial headroom require predictor-response information and enough decision margin for the router to distinguish candidates.

The multi-target study shows why both stages matter. Structural headroom is positive on every evaluated target, but learned routing is more reliable on some targets than others.

METR-LA has the widest target-level variation. This pattern matches the mechanism analysis: structural opportunity alone does not determine how much value a router can realize. Predictor-response features and the utility gap between the best candidates are the operating quantities.

The main limitations are computational and structural. R-MUR is greedy and can miss pure complementarity, where every individual candidate has nonpositive utility but a bundle is useful. Its labels require counterfactual evaluations of the fixed predictor, so offline label cost is part of the system cost. The calibration-derived stopping threshold controls the gain-harm trade-off, and its guarantee is per-decision.

A. Structural headroom is not the same as realizable headroom

The strong-backbone study separates two quantities that the earlier results could not distinguish. Structural headroom is a property of the predictor and the candidate pool: with the subset-capable transformer, the sequential oracle still beats the standalone oracle by .0148 absolute on METR-LA, so state conditioning continues to create value. Realizable headroom is a property of the information available to the router. With the ridge expert the learned router converted a substantial share of the structural opportunity into gain; with the nonlinear expert the response-free routers are statistically indistinguishable from random selection even though the same structural opportunity is present.

The general-loss analysis of Section V explains the mechanism. The expectation of the exact identity conditions on the state as $\mathbb{E}[m(j | A) | x, A] = \langle g_A(x), d_{A,j} \rangle - \|d_{A,j}\|^2 - \lambda c_j$ with $g_A(x) = 2(\mu(x) - f_A(x))$. The second term is known as soon as the predictor has been probed, but the first requires predicting the residual of the *better* predictor. A stronger predictor has already absorbed the systematic part of that residual, so what remains is closer to noise, and the expected

TABLE XV
WHERE THE FRAMEWORK PAYS OFF. Δ STATIC IS R-MUR AT $q = 4$
AGAINST STANDALONE UTILITY WITH A TARGET-LEVEL INTERVAL.

Setting	Headroom	Δ Static
Traffic, ridge (32 targets)	.09-.11	+0.0122 [+0.010, +0.015]
Traffic, ridge (PEMS-BAY)	.11	+0.0288 [+0.021, +0.040]
Traffic, nonlinear	.15	+0.0028 [-0.001, +0.008]
Traffic, nonlinear, unadapted	.16	+0.0122 [+0.006, +0.020]
PEMS-BAY, nonlinear	.15	-.0008 [-0.004, +0.002]
Demos, in-context	$\leq 5 \cdot 10^{-4}$	3/5 benchmarks
Demos, forced budget	$\leq 5 \cdot 10^{-4}$	5/5 benchmarks
Demos, redundant pools	$\leq 9 \cdot 10^{-4}$	1/5; 5/5 forced

ranking is increasingly governed by the response penalty alone. This is a boundary condition of utility routing rather than a defect of a particular router: as fixed predictors improve, context selection has less ex-ante identifiable value, even while its oracle value persists. Practically, the diagnostic to run before deploying a router is the one used here: compare the oracle ladder against the best response-free ranking and check whether the cached view correlates with the true marginal at all. Both comparisons are computed from the deployment’s own protocol and cost one sweep each. Fitting a predictability probe is deliberately *not* on this list: as Section VI-E shows, even at 2,000 training episodes the probe cannot separate a weak expert from a strong one, so it can rank geometries but not cells.

B. Where sequential routing pays off

Table XV collects the settings evaluated in this paper. Two conditions must hold together. The predictor must leave a systematic residual for contexts to correct: the ridge expert does, and the learned router converts a double-digit share of the standalone oracle into gain with intervals well above zero. The candidate set must make a candidate’s value depend on what is already selected: constructed redundant traffic pools do, whereas demonstration pools without redundancy have a flat state-conditioned ceiling ($H_{\text{state}} \leq 5 \times 10^{-4}$). When the predictor is strong and well calibrated, the first condition weakens even though the oracle gap remains, and no evaluated method — utility-based or policy-gradient — recovers it. When the pools are not redundant, the second condition fails and nearest-neighbour relevance is within a fraction of a percent of the oracle. The stopping rule is a separate design axis: forcing the budget to be spent turns a 3-of-5 result into 5-of-5 in the second family without changing the ranking.

Two further axes decide whether routing is worth attempting, and both are counter-intuitive. The first is adaptation. Fine-tuning the fixed predictor on the deployment is normally free accuracy, and it is exactly the operation that destroys routing value: in a controlled sweep of adaptation epochs, two epochs improve anchor error by 7.1% while removing about five sixths of the share of the oracle the router could realise, and the structural opportunity is unchanged throughout (+.0831 to +.0858). Twenty-eight further epochs buy a further 1.2% of accuracy and no identifiable value at all. The practical reading is that routing and heavy test-time adaptation are

substitutes: the residual a router exploits is the same residual adaptation absorbs, so a practitioner should decide which of the two is to capture it rather than expecting both. The second axis is the size of the candidate library. Enlarging the pool from sixteen to one hundred and twenty-eight candidates raises the standalone-oracle opportunity by forty percent and drives the router to the level of random selection, because the value that appears is value nothing can identify in advance. Retrieving from a bigger library therefore makes this problem harder, not easier — the opposite of the usual scaling intuition — and the useful move at large pool sizes is to make the pool more redundant or the predictor less adapted, not larger.

C. When a simple baseline is hard to beat

The second task family shows the same boundary from the other side. Because the demonstration pools are not redundant, the sequential and standalone oracles coincide and a nearest-neighbour relevance rule reaches 99.4–99.9% of the oracle gain. In that regime the ranking problem is essentially solved by similarity, and no learned router, including R-MUR, improves on it uniformly. The framework is therefore best applied where candidates overlap in what they contribute: redundancy is what makes the value of a candidate depend on what has already been selected, and that dependence is what a sequential router can exploit.

VIII. CONCLUSION

We formulated context selection through the set-conditioned marginal reduction in end-task loss. This decision object exposes structural headroom created by state conditioning, and it separates that opportunity from the accuracy with which a router can estimate it. R-MUR combines cached screening with compact predictor-response summaries and a gap-weighted ranking objective. Across six standard traffic benchmarks, R-MUR improves learned context selection over standalone and cached baselines, and a multi-target study shows that structural headroom persists across sensors.

Two additions sharpen the scope of that claim. First, a general smooth-loss analysis shows that marginal utility decomposes into the alignment between the predictor’s response and its residual plus a curvature penalty, with cross-entropy and other Bregman losses covered by exact or two-sided statements; predictor response is therefore the theoretically matched observation, and the numerical study confirms the bound empirically. Second, a subset-capable nonlinear backbone preserves structural headroom but reduces the share of it that a router can realize from observable features: the learned response-free routers become indistinguishable from random selection and R-MUR’s advantage over them is no longer statistically resolved, while a second task family whose candidates are not redundant is already solved by nearest-neighbour relevance. Predictor response and decision margin remain the key quantities for turning structural opportunity into learned routing gains, and the strength of the fixed predictor determines how much of that opportunity is identifiable before the decision is made.

APPENDIX

A. Proof of Proposition 1

Apply Taylor’s theorem with Lagrange remainder to $t \mapsto \ell(f_A(x) + t d_{A,j}(x), y)$ on $[0, 1]$. There exists $\xi = \xi(x, y)$ on the segment with

$$\ell(f_{A \cup \{j\}}, y) = \ell(f_A, y) + \nabla \ell(f_A, y)^\top d_{A,j} + \frac{1}{2} d_{A,j}^\top \nabla^2 \ell(\xi, y) d_{A,j}. \quad (24)$$

Rearranging, subtracting λc_j , and taking expectations gives Equation (17). Under the L -smoothness assumption, $|d^\top \nabla^2 \ell(\xi, y) d| \leq L \|d\|^2$ for every realisation, which yields Equation (18) after taking expectations.

B. Proof of Proposition 2

The Bregman three-point identity states that for any u, v, w in the domain of ϕ ,

$$D_\phi(u, v) - D_\phi(u, w) = (\nabla \phi(w) - \nabla \phi(v))^\top (u - v) - D_\phi(v, w). \quad (25)$$

Both sides expand to $\phi(u) - \phi(v) - \langle \nabla \phi(v), u - v \rangle$ minus $\phi(u) - \phi(w) - \langle \nabla \phi(w), u - w \rangle$; all ϕ terms cancel. Set $u = y$, $v = f_A$, $w = f_{A \cup \{j\}} = f_A + d_{A,j}$, take expectations, and subtract λc_j to obtain Equation (19). The final claim follows from the standard strong-convexity and smoothness bounds on Bregman divergences.

C. Proof of Proposition 3

For $\ell(f, y) = -f_y + \log \sum_k e^{f_k}$ we have $\nabla \ell(f, y) = p - e_y$ and $\nabla^2 \ell(f, y) = \text{diag}(p) - pp^\top \succeq 0$. Since $(\text{diag}(p) - pp^\top) \mathbf{1} = 0$, the quadratic form is invariant to adding a constant to v , so $v^\top (\text{diag}(p) - pp^\top) v = \text{Var}_p(v)$ for the random variable taking value v_k with probability p_k . Let $M = \max_k v_k$, $m = \min_k v_k$, and $\mu_p = \sum_k p_k v_k$. The Bhatia–Davis inequality gives $\text{Var}_p(v) \leq (M - \mu_p)(\mu_p - m) \leq (M - m)^2/4$. Replacing v by $v - \bar{v} \mathbf{1}$ (which leaves both the quadratic form and the range bound valid) makes the vector zero-sum, so $M \geq 0 \geq m$ and $\|v\|^2 \geq M^2 + m^2$. Hence $(M - m)^2 = (M + |m|)^2 \leq 2(M^2 + m^2) \leq 2\|v\|^2$ and therefore $v^\top (\text{diag}(p) - pp^\top) v \leq \frac{1}{2} \|v\|^2$. The bound is attained at $p = (\frac{1}{2}, \frac{1}{2}, 0, \dots)$ and $v = (1, -1, 0, \dots)/\sqrt{2}$, so the constant $\frac{1}{2}$ is tight. Applying Proposition 1 with $L = \frac{1}{2}$ and using $-\nabla \ell(f_A, y) = e_y - p_A$ gives Equation (20); the upper bound holds because the remainder $-\frac{1}{2} d^\top H d$ is non-positive for $H \succeq 0$.

D. Proof of Proposition 4

For squared loss the exact identity of Proposition 2 reads $\tilde{m}(j | A) = \ell(f_A, y) - \ell(f_{A \cup \{j\}}, y) - \lambda c_j = 2d_{A,j}^\top (y - f_A) - \|d_{A,j}\|^2 - \lambda c_j$. The response $d_{A,j} = d_{A,j}(x)$ is a function of the episode input, so it is measurable with respect to (x, A) and can be taken outside the conditional expectation; the tower property gives Equation (21) with $\mu(x) = \mathbb{E}[y | x, A]$. Linearity in $d_{A,j}$ and the response-only quadratic penalty follow immediately. For a general differentiable loss the same argument leaves $\nabla \ell(f_A, y)$ inside the expectation, and this gradient depends on y beyond the residual, which is why state features are also required.

TABLE XVI

NUMERICAL VALIDATION OF THE RESPONSE–UTILITY RELATIONS. ALL NUMBERS ARE PRODUCED BY THE RELEASED SCRIPTS; THE CROSS-ENTROPY CHECK USES THE FROZEN IN-CONTEXT CLASSIFIER OF SECTION VI-F.

Check	Object	Result
Squared-loss identity	forecast expert	max. deviation 3.3×10^{-7}
CE sandwich (Equation (20))	in-context classifier	0 violations / 92,995
First-order vs. marginal	in-context classifier	Spearman .971, R^2 .853
Response vs. state only	forecast expert	R^2 gain +0.67
Hessian spectral bound	softmax, all $C \leq 50$	maximum = 0.5

E. Numerical validation

We validate the identities numerically rather than relying on the derivations alone.

a) *Squared loss (exactness)*.: Using the subset-capable nonlinear forecasting expert of Section VI-E on held-out episodes, we compared the exact marginal $\ell(f_A, y) - \ell(f_{A \cup \{j\}}, y)$ with the right-hand side of the squared-loss identity, and separately verified that the seven response features of the router reproduce the response statistics. Over 384 held-out (state, candidate) pairs the maximum absolute deviation is 3.3×10^{-7} in normalised units, i.e. floating-point exactness. Unit tests enforce the same identity and the inertness of unselected candidates.

b) *Cross-entropy (two-sided bound)*.: For the frozen in-context classifier of the second task family we collected 92,995 (state, candidate) samples together with 2,975,840 per-query rows and evaluated the sandwich of Proposition 3. The bound held in *every* sample at both aggregation levels (smallest upper slack 3.0×10^{-8} , smallest lower slack 1.0×10^{-3}). The first-order term tracks the true marginal closely: Spearman correlation .971 and $R^2 = .853$ pooled, with per-benchmark Spearman between .970 and .993. The bound is therefore not merely valid but informative, and the empirical tightness $\rho = (m - \text{first})/(-\frac{1}{4} \|d\|^2) \in [0, 1]$ is reported in the released artifact.

c) *Response sufficiency*.: For the squared-loss expert we fitted the same ridge head twice on identical held-out state samples: once with state features only and once with state plus response features. The response features raise the out-of-sample R^2 from -0.89 to -0.22 on a held-out episode split (a gain of 0.67), and the cross-entropy family shows the same direction with a Spearman correlation of .97 between the first-order term and the realised marginal. Both are the practical content of Proposition 4: the response supplies the interaction term that a state-only score cannot represent.

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